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Roll No

ME-701(C) (GS)

B.E. VII Semester Examination, June 2020

Grading System (GS)

Industrial Robotics

Time : Three Hours

Maximum Marks : 70

Note : i) Attempt any five questions.

ii) All questions carry equal marks.

1. What is Jacobean of a robot system? Derive the Jacobean matrix for the question stated above.
2. Write down the difference between the serial robot and parallel robots.
3. Define the terms:
 - i) Resolution
 - ii) Accuracy
 - iii) Repeatability
 - iv) Work envelope
4. Define the following:
 - a) Load carrying capacity
 - b) Work volume
 - c) End - effector
5. Define “Degree of freedom” of a rigid body in space calculate the degree of freedom of a GYROSCOPE.
6. With neat sketches differentiate and highlight the four common types of robot configurations.
OR
Explain how the performance of robotic system is studied?
7. Explain in brief the image reorganization methods.
8. Answer any four of the following:
 - a) What are the four D’s of robotics?
 - b) What is the purpose of wrist? What is its degree of freedom?
 - c) What are the essential components of the sensor?
 - d) What are the advantages and disadvantages of on-line programming?
 - e) Enumerate the factors influences the robot operational cost in detail.
 - f) Enumerate the factors influencing direct and indirect revenue generation by utilizing robots in detail.
